

CHAPTER 4: MOTHERBOARD AND COMPUTER PORTS

Several thin, white, parallel lines are drawn diagonally across the right side of the image, starting from the top right and extending towards the bottom left.

INTENDED LEARNING OUTCOMES (ILOS)

AT THE END OF THIS LESSON, THE STUDENT IS EXPECTED TO:

- ❑ LEARN EVERY PART OF THE MOTHERBOARD AS WELL AS THEIR FUNCTIONS;
- ❑ LEARN THE DIFFERENT KINDS AND TYPES OF PORT AND THEIR FUNCTIONS;
- ❑ ENUMERATE THE MAJOR MOTHERBOARD COMPONENTS AND THEIR FUNCTIONS;
- ❑ UNDERSTAND BASIC INPUT/OUTPUT SYSTEM;
- ❑ UNDERSTAND THE INFORMATION ABOUT CMOS RAM;
- ❑ BE FAMILIAR WITH HOW THE CACHE MEMORY WORKS;

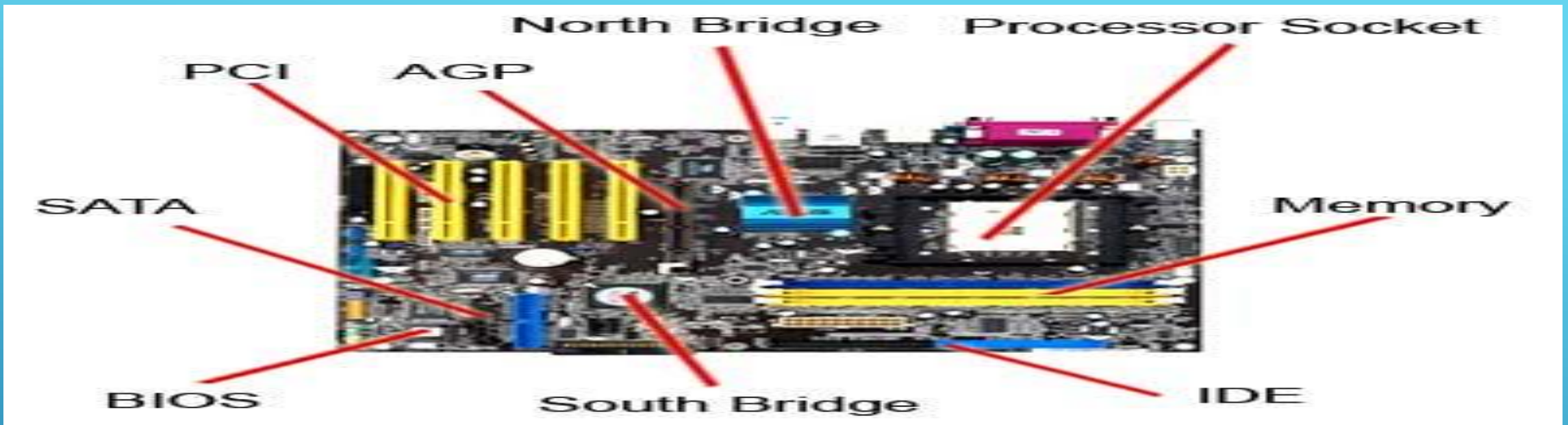
INTENDED LEARNING OUTCOMES (ILOS)

AT THE END OF THIS LESSON, THE STUDENT IS EXPECTED TO:

- ❑ EXPLAIN THE FUNCTIONS OF THE EXPANSION BUS;
- ❑ LEARN WHAT IS THE USE OF CHIPSETS OF A COMPUTER;
- ❑ UNDERSTAND THE FUNCTION OF A CPU CLOCK;
- ❑ ENUMERATE THE SWITCHES AND JUMPERS OF A MOTHERBOARD;
- ❑ LIST AND UNDERSTAND ALL THE DIFFERENT TYPE OF PORTS

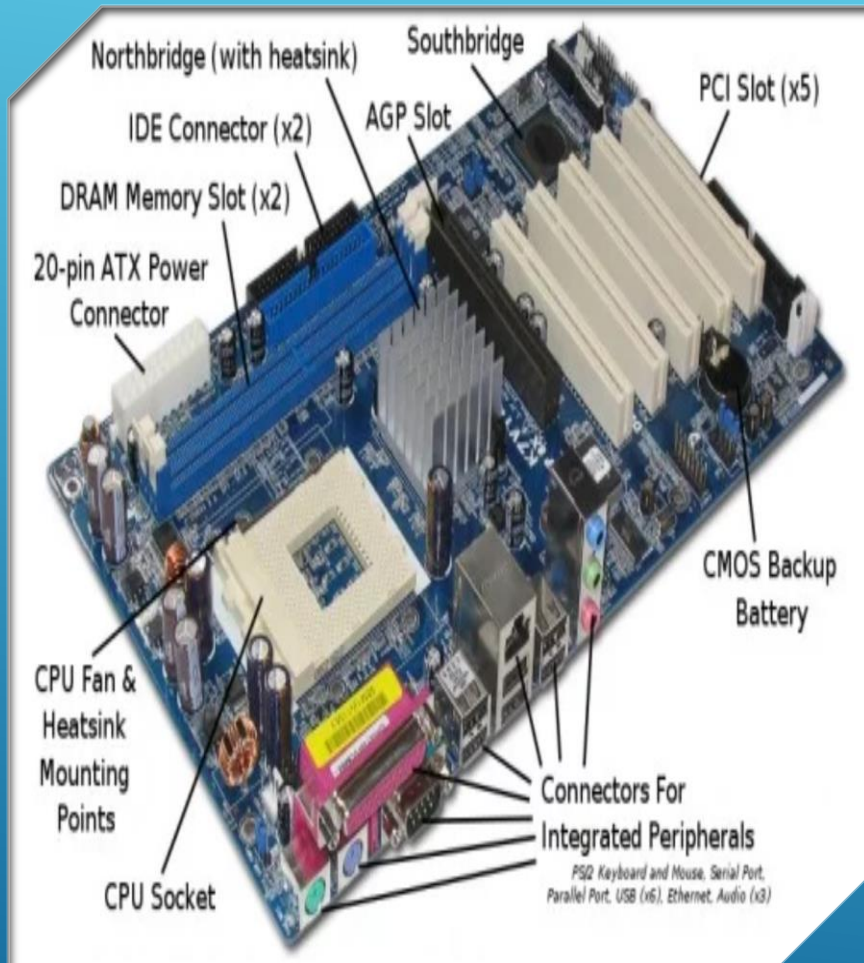
Mother Board

Mother Board (known as the Mainboard, system board, base board, planar board or logic board, or colloquially, a mobo). It holds and allows communication between many of the crucial electronic components of a system, such as the central processing unit (CPU) and memory, and provides connectors for other peripherals. Motherboard contains significant sub-systems such as the central processor, the chipset's input/output and memory controllers, interface connectors, and other peripheral devices.



COMPUTER PORTS:

A COMPUTER PORT IS A SPECIALIZED OUTLET ON A PIECE OF EQUIPMENT TO WHICH A PLUG OR CABLE CONNECTS AND PORTS SERVE AS LINKAGE BETWEEN THE COMPUTER TO PERIPHERAL DEVICES AND OTHER COMPUTERS.



SOME OF THE MAJOR COMPONENTS OF A MOTHERBOARD:

Some of the Major Components of a Motherboard:

Major components of motherboard Processor (CPU), memory (RAM), and expansion slots attached in the mother board and very important for the computer to function.

CENTRAL PROCESSING UNIT (CPU)

The CPU (Known as the microprocessor or the processor) is serve as the computer's brain and responsible for computations, fetching, decoding and executing program instructions.

PROCESSOR CHIP

The processor chip is identified by the processor type and the manufacturer. The information is from the manufacturer which is identify the processor that fits in the sockets.





RANDOM ACCESS MEMORY (RAM)

Random Access Memory or RAM that stores dynamic data temporarily and the working place of the computer. The RAM serve place where data and active programs occur and it loses the contents once power is turned off or shutdown.

BASIC INPUT/OUTPUT SYSTEMS (BIOS)

The BIOS have all the codes required for controlling the keyboard. Displaying the screen, disk drives, serial communications, and a number of miscellaneous functions. BIOS is the link in the between the software and computer hardware.

The System BIOS is a ROM chip on the mother board used during the startup routine (boot process) to check out the systems and prepare to run the hardware. The BIOS is stored on a ROM chip because ROM retains information even when no power is being supplied to the computer.



Motherboard also include a small separate block of memory made from CMOS RAM chips which is kept alive by a battery (known as a CMOS battery) even when the PC's power is off. This prevents reconfiguration when the PC is powered on.

CMOS devices require very little power to operate.

The CMOS RAM is used to store basic Information about the PC's configuration for instances:

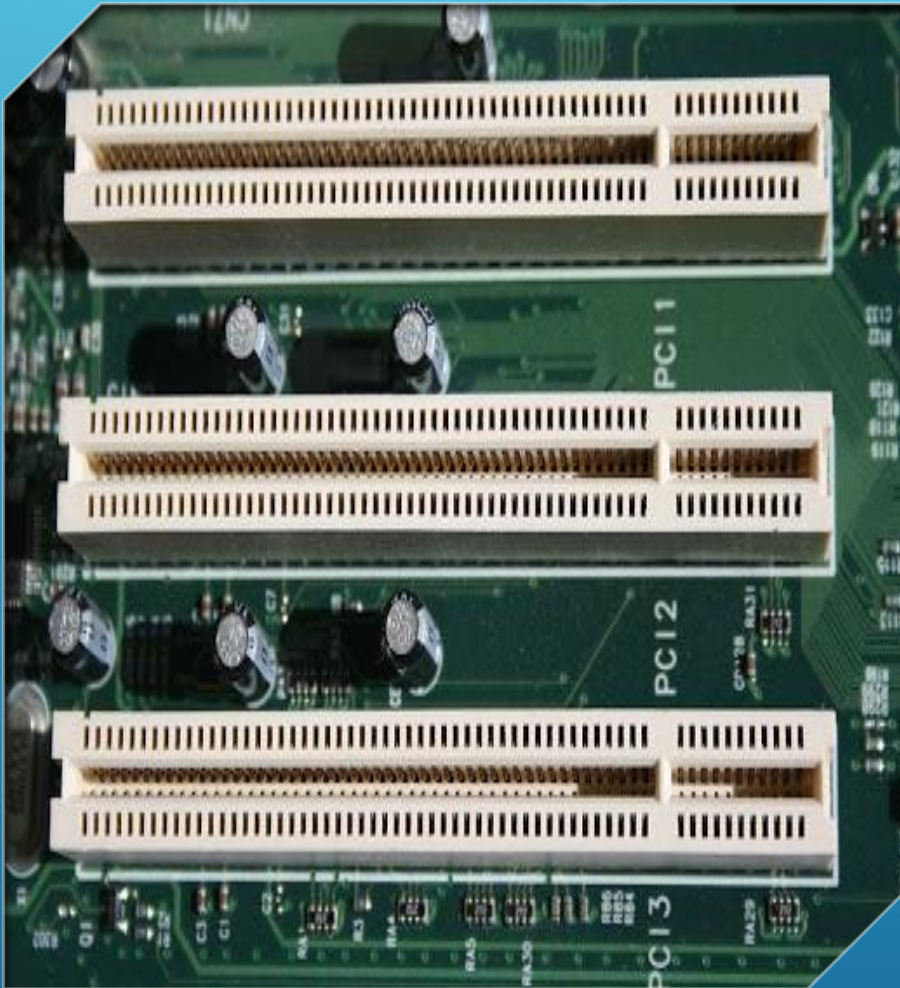
- RAM size
- Information about CPU
- Serial and parallel port information
- Floppy disk and hard disk drive types
- Power Saving settings
- Plug and Play information
- Date and Time



CACHE MEMORY

A small block of high-speed memory that boosts the PC's performance by pre-loading information from the (relatively slow) main memory and passing it to the processor on demand.

Every CPU has a built-in cache and called as the Level 1 or the primary cache memory. This is supplemented by an external cache memory called the secondary cache or level 2 cache. If there is a cache implemented outside the die, which will be called the level 3 cache.



EXPANSION BUS

Expansion bus is an input/output pathway to the peripheral devices coming from the CPU. It is typically made up of a peripheral devices coming from the CPU and signals such as data, memory, addresses, and power and control signals are carried from component to component by the busses. Other types of busses are ISA and EISA.

The Function of expansion buses is it enhances the PC's capabilities by slotting cards into the expansion slots, which allows the user to add missing features into computer.

CHIPSET

A chipset is a group of tiny circuits that coordinates the data flow back and forth from a computer's key components.

A chip is also the one that controls data flow to and from the hard disk and the devices that are connected to the channels of IDE.

The 2 Main Chipsets of a Computer:

- The North Bridge (memory controller) controls the transfer between the RAM and the processor. Sometimes it is called as the GMCH (Graphics and Memory Control Hub)
- The South Bridge (expansion controller) handles the communication between slower peripheral devices. The term “bridge” is generally used for designating a component which connects two buses.

CPU clock synchronizes the operation of all parts of the PC and provides the basic timing signal for the CPU. Using a quartz crystal the CPU clock breathes life into the microprocessor by feeding it a constant flow of pulses.

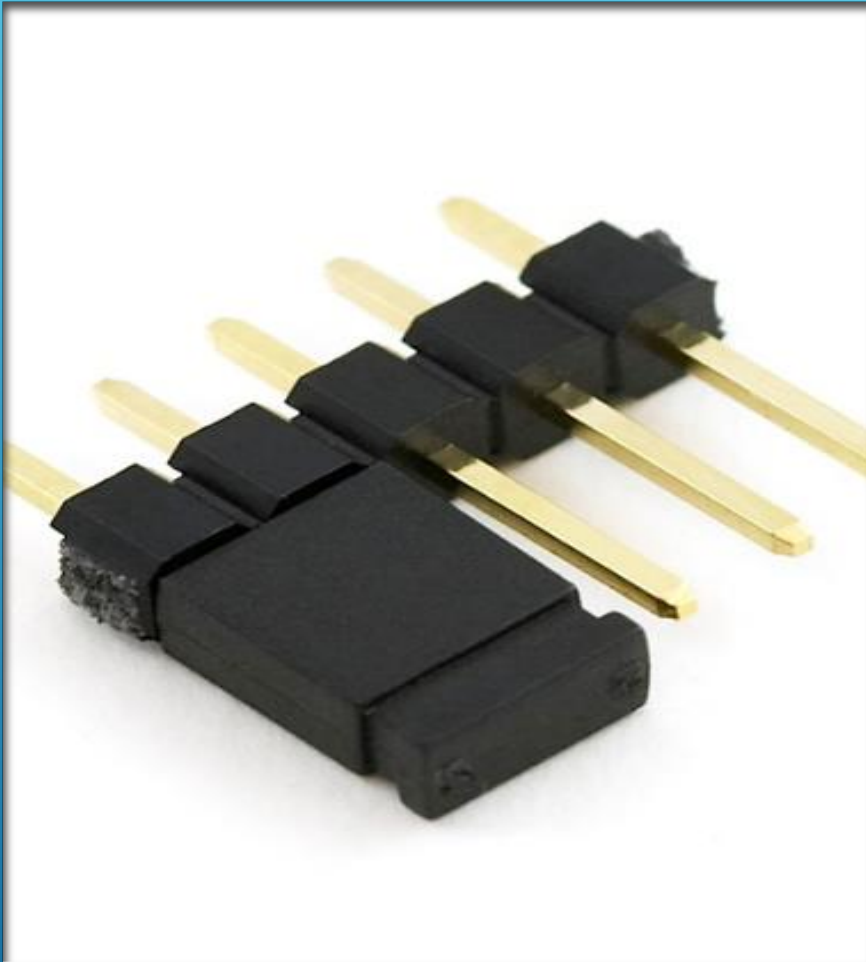
Real time clock or System clock keeps track of the time of day and makes this data available to the software.

Time sharing clock interrupts the CPU at regular intervals and allows the operating system to divide its time between active users and/or applications.



SWITCHES AND JUMPERS

- DIP (Dual In-Line Package) switches are small electronic switches found on the circuit board that can be turned on or off just like a normal switch.



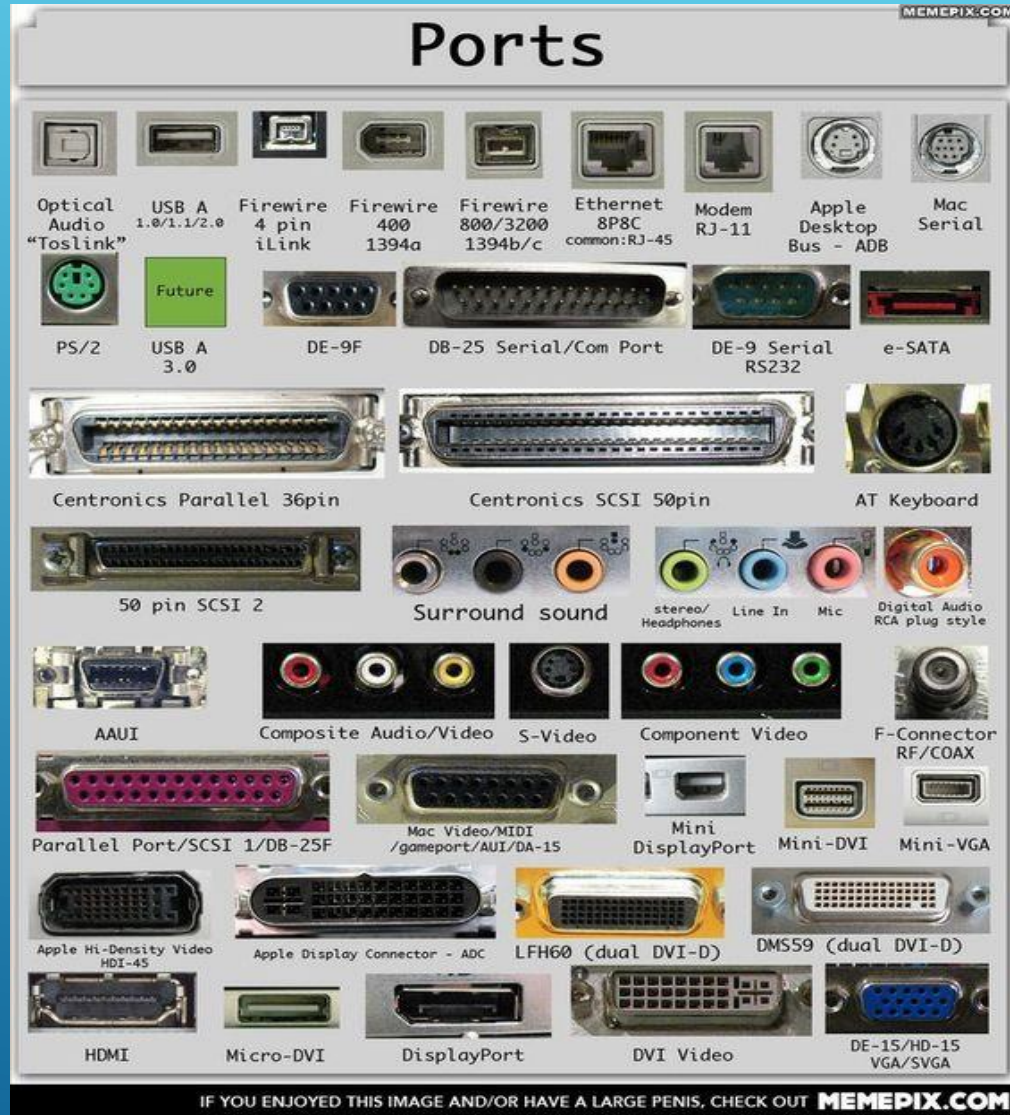
SWITCHES AND JUMPERS

- Jumper pins are small protruding pins on the mother board. A jumper cap or bridge is used to connect or short a pair of jumper pins.



SWITCHES AND JUMPERS

- Jumper caps are metal that close an electrical circuit and sometimes used to configure expansion boards. By placing a jumper plug over a different set of pins, you can change a board's parameters.



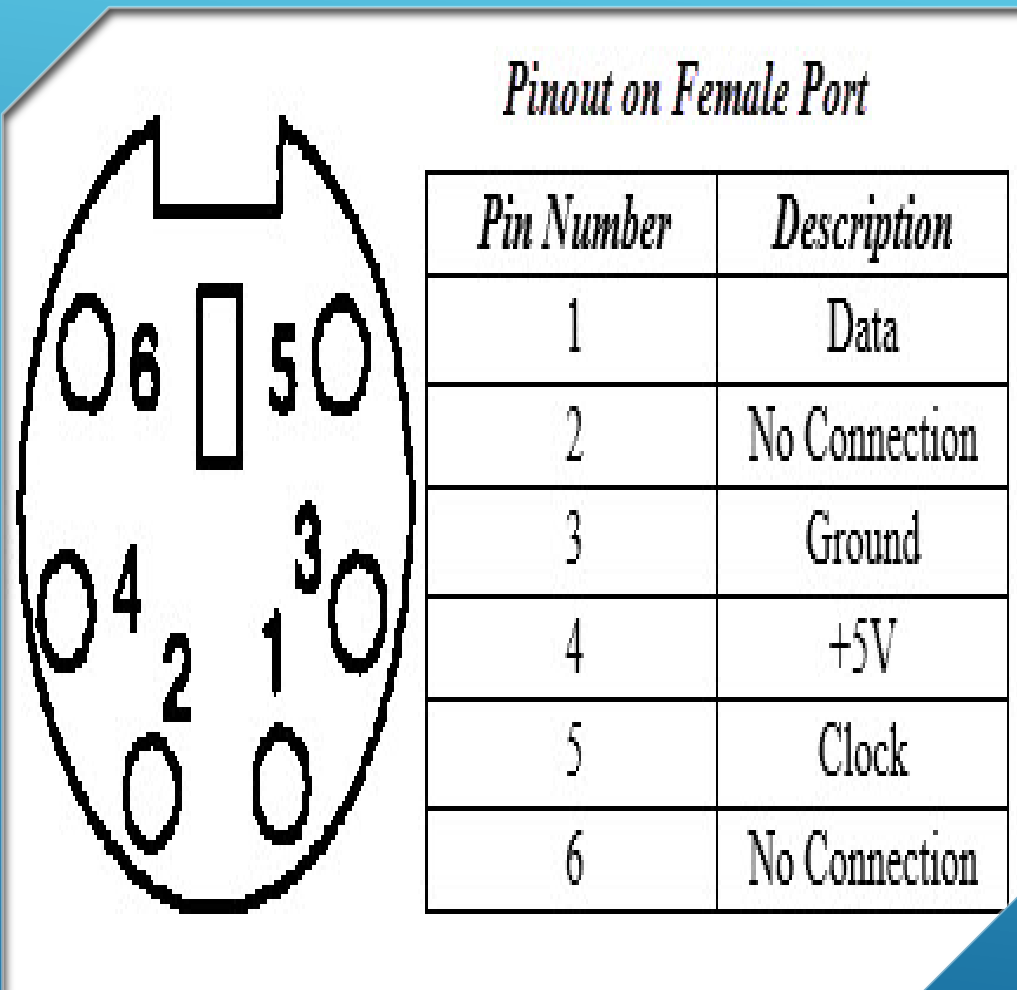
COMPUTER PORT

Computer port is also called as a Communication Port as it is responsible for communication between the computer and its peripheral device. Communication ports can be divided into two types based on the type or protocol used for communication. They are Serial Ports and Parallel Ports.

COMPUTER PORT

Serial Port is an interface through which peripherals can be connected using a serial protocol which involves the transmission of data one bit at a time over a single communication line. Common types of serial port is a D-Subminiature or a D-sub connector that carry RS-232.

Parallel port is an interface through which the communication between a computer and its peripheral device is in a parallel manner i.e. data is transferred in or out in parallel using more than one communication line or wire. Printer port is an example of parallel PS/2.



PS/2 CONNECTOR

PS/2 connector is developed by IBM for connecting mouse and keyboard. P/S2 connectors are color coded as purple for keyboard and green for mouse and is has a 6-pin DIN Connector.

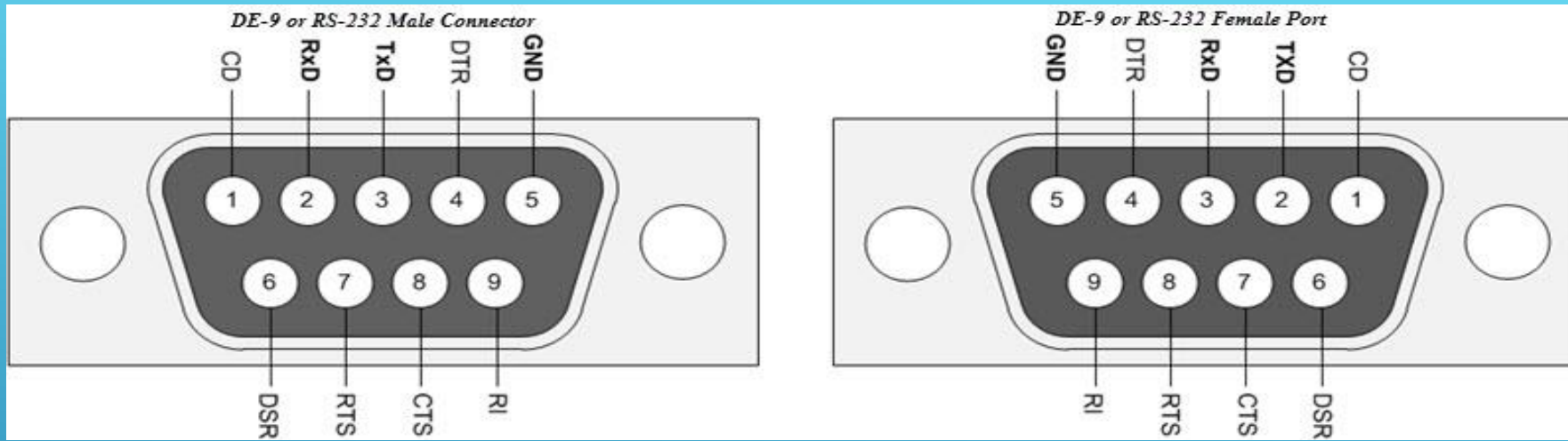
This is a pinout diagram of PS/2 female connector even though the pinout of both mouse and keyboard PS/2 ports same, computers do not recognize the device when connected to wrong port.

Serial Port

The term Serial Port is used to refer the interface that is compliant to RS-232 standard. There are two types of serial ports that are commonly found on a computer: DB-25 and DE-9.

DB-25

Db-25 is a variant of D-sub connector and is the original port for RS-232 serial communication and rarely used as a serial port and often used as a parallel printer port as a replacement of the Centronics Parallel 36 pin connector.



DE-9 OR RS-232 OR COM PORT

A DE-9 port is also called as a COM port and allows full duplex serial communication between the computer and its peripheral. Used as serial interface with mouse, keyboard, modem, uninterruptible power supplies (UPS) and other external RS-232 compatible devices.



PARALLEL PORT OR CENTRONICS 36 PIN PORT

Parallel port is an interface between computer and peripheral devices like printers with parallel communication. The Centronics ports is a 36-pin port that was developed as an interface for printers and scanners and hence a parallel ports is also called as a Centronics port.

AUDIO PORTS

Audio Ports

Audio ports are used to connect speakers or other audio output devices with the computer.

Surround Sound Connectors or 3.5 mm TRS Connector

A 6-connector system included on majority of computers for audio out as well as a microphone connection. These 6 connectors can be used for a surround sound configurations of up to 8 channels.



Port	2-Channel	4-Channel	6-Channel	8-Channel
Blue	Line In	Line In	Line In	Line In
Green	Line Out	Front Speakers	Front Speakers	Front Speakers
Pink	Mic In	Mic In	Mic In	Mic In
Orange			Center/Subwoofer	Center/Subwoofer
Black		Rear Speakers	Rear Speakers	Rear Speakers
Grey				Side Speakers



S/PDIF/TOSLINK

The Sony /Philips Digital Interface Format (S/PDIF) is an audio interconnect used in home media. It supports digital audio and can be transmitted using coaxial RCA Audio cable or an optical fiber TOSLINK connector. TOSLINK (Toshiba Link) is most frequently used digital audio port that can support 7.1 channel surround sound with just one cable.



VGA PORT

VGA (Video Graphics Array) it is a D-sub connector consisting of 15 pins in 3 rows. The connector is called as DE-15. It is the main interface between computers and older CRT monitors.

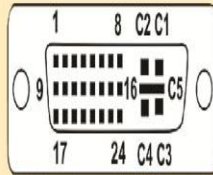


DIGITAL VIDEO INTERFACE (DVI)

DVI is a high speed digital interface between a display controller like a computer and a display device like a monitor. IT was developed with an aim of transmitting a lossless digital video signals and replace by the analogue VGA technology.

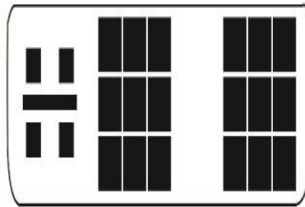
There are three types of DVI connectors based on signal it can carry: DVI-I, DVI-D and DVI-A. DVI-I is a port with integrated analogue and digital signals. DVI-D supports only digital signals and DVI-A supports only analogue signals.

DVI

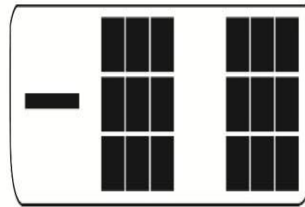


Mating face of DVI female

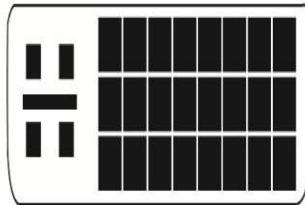
PIN#	SIGNAL	PIN#	SIGNAL
1	T.M.D.S DATA 2 -	16	HOT PLUG DETECT
2	T.M.D.S DATA 2 +	17	T.M.D.S DATA 0 -
3	T.M.D.S DATA 2/4 SHIELD	18	T.M.D.S DATA 0 +
4	T.M.D.S DATA 4 -	19	T.M.D.S DATA 0/5 SHIELD
5	T.M.D.S DATA 4 +	20	T.M.D.S DATA 5 -
6	DDC CLOCK	21	T.M.D.S DATA 5 +
7	DDC DATA	22	T.M.D.S CLOCK SHIELD
8	ANALOG VERT. SYNC	23	T.M.D.S CLOCK +
9	T.M.D.S DATA 1 -	24	T.M.D.S CLOCK -
10	T.M.D.S DATA 1 +		
11	T.M.D.S DATA 1/3 SHIELD	C1	ANALOG RED
12	T.M.D.S DATA 3 -	C2	ANALOG GREEN
13	T.M.D.S DATA 3 +	C3	ANALOG BLUE
14	+5V POWER	C4	ANALOG HORIZ. SYNC
15	GND	C5	ANALOG GROUND



DVI-I (Single Link)



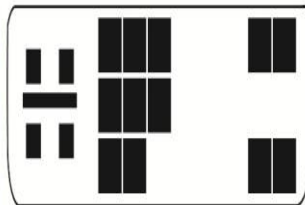
DVI-D (Single Link)



DVI-I (Dual Link)



DVI-D (Dual Link)



DVI-A (Analog)

DIGITAL VIDEO INTERFACE (DVI)

The digital signals can be either single link or dual link where a single link supports a digital signal up to 1920X1080 resolution and a dual link supports a digital signal up to 2560X1600 resolution. The following image compares the structure of DVI-I, DVI-D and DVI-A types along with the pinouts.



MINI DVI

Mini-DVI port is developed by Apple as an alternative to Mini-VGA port and is physically similar to one. It is smaller than regular DVI port. It is a 32-pin port and is capable of transmitting DVI, composite, S-Video and VGA signals with respective adapters.



MICRO-DVI

Micro-DVI port, as the name suggests is physically smaller than Mini-DVI and is capable of transmitting only digital signals. This port can be connected to external devices with DVI and VGA interface and respective adapters are required.

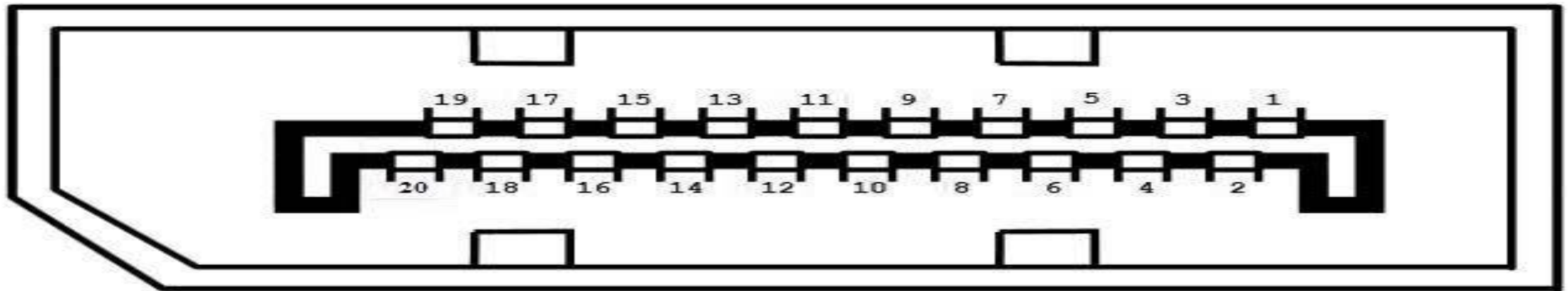


DISPLAY PORT

Display Port is a digital display interface with optimal multiple channel audio and other forms of data. Display Port is developed with an aim of replacing VGA and DVI ports as the main interface between monitor.

Display Port has a 20-pin connector, which is very less number when compared to DVI port and offers better resolution.

DISPLAY PORT



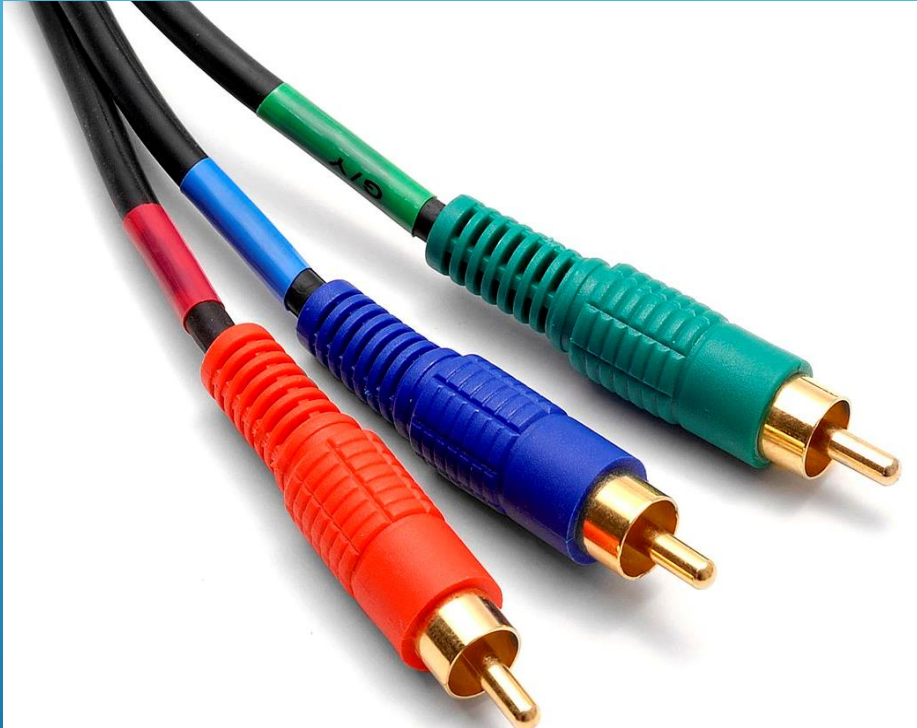
<i>Pin NO</i>	<i>Pin Name</i>	<i>Description</i>
Pin 1	ML_Lane 0 (p)	Lane 0 (positive)
Pin 2	GND	Ground
Pin 3	ML_Lane 0 (n)	Lane 0 (negative)
Pin 4	ML_Lane 1 (p)	Lane 1 (positive)
Pin 5	GND	Ground
Pin 6	ML_Lane 1 (n)	Lane 1 (negative)
Pin 7	ML_Lane 2 (p)	Lane 2 (positive)
Pin 8	GND	Ground
Pin 9	ML_Lane 2 (n)	Lane 2 (negative)
Pin 10	ML_Lane 3 (p)	Lane 3 (positive)

<i>Pin NO</i>	<i>Pin Name</i>	<i>Description</i>
Pin 11	GND	Ground
Pin 12	ML_Lane 3 (n)	Lane 3 (negative)
Pin 13	CONFIG1	connected to Ground1)
Pin 14	CONFIG2	connected to Ground1)
Pin 15	AUX CH (p)	Auxiliary Channel (positive)
Pin 16	GND	Ground
Pin 17	AUX CH (n)	Auxiliary Channel (negative)
Pin 18	Hot Plug	Hot Plug Detect
Pin 19	Return	Return for Power
Pin 20	DP_PWR	Power for connector (3.3 V 500 mA)



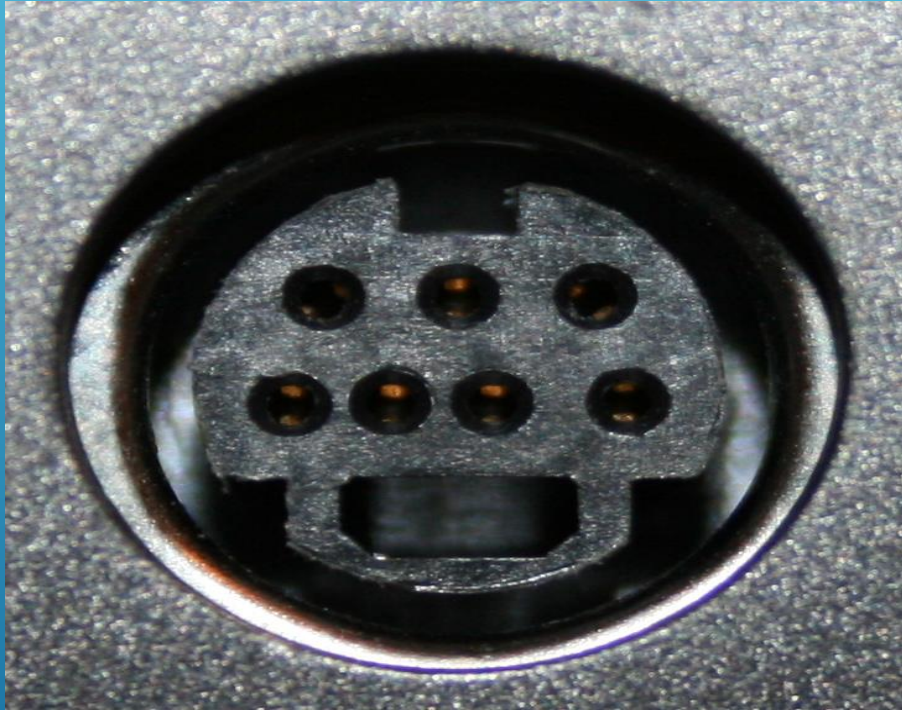
RCA CONNECTOR

RCA Connector can carry composite video and stereo audio signals over three cables. Composite or yellow colored RCA connector that transmits analogue video signals. The red and white connectors are used for stereo audio signals (red for right channels and white for left channel).



COMPONENT VIDEO

Component Video is an interface where the video signals are split into more than two channels and the quality of the video signals is better than composite video and can transmit both analogue and digital video signals. The port of commonly found Component video uses 3 connectors and it is color coded as Green, Blue and Red.



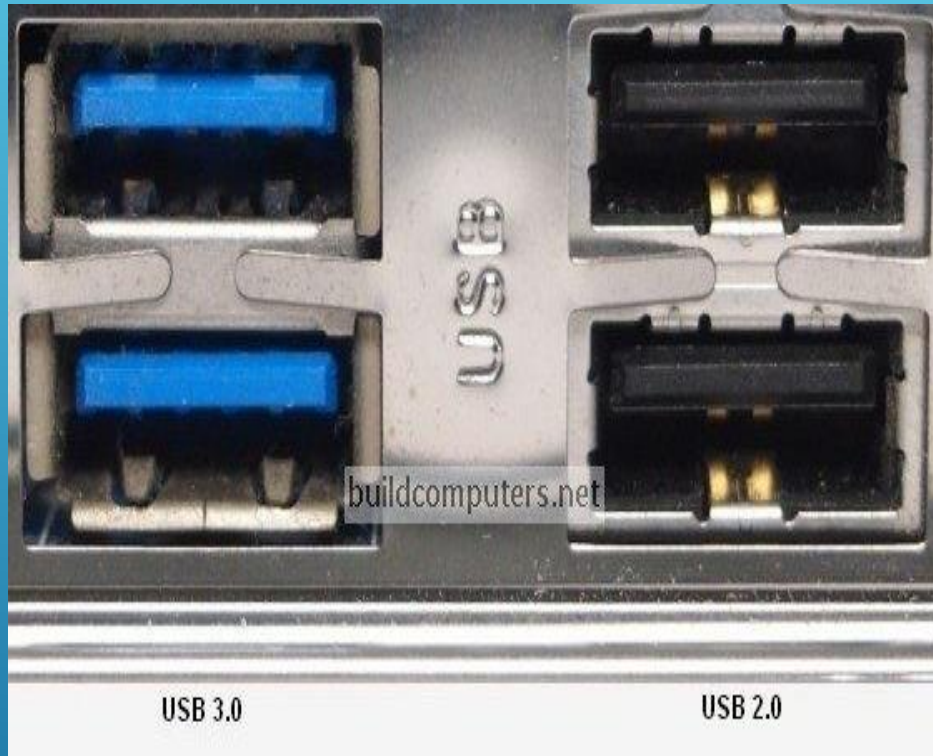
S-VIDEO

S-Video or separate Video connector is used for transmitting only video signals. The picture quality is better than that of composite. Out of the 4 pins, one pin is used to carry the intensity signals (black and white) and other pin is used carry color signals.



HDMI

HDMI is an abbreviation of high Definition Media Interface. HDMI is a digital interface to connect High Definition and Ultra High Definition devices like Computer monitors, HDTVs, Blu-Ray players, gaming consoles, High Definition Cameras. The HDMI connector consists of 19 pins and the latest versions HDMI i.e. HDMI 2.0 can carry digital video signal up to a resolution of 4096X2160 and 32 audio channels.



USB C
1 = +5 volts
2 = - Data
3 = + Data
4 = Ground

USB

Universal Serial Bus (USB) replaced serial ports, parallel ports PS/2 connectors, game ports and power charges for portable devices. There are three kinds of USB ports: Type A, Type B or mini USB and Micro USB.

USB Type A

USB Type-A port is a 4-pin connector. There are different versions of type – A USB ports: USB 1.1, USB 2.0 and USB 3.0. USB 3.0 is the common standard and supports a data rate of 480MBps.

USB 3.1 is also released and supports a data rate up to 10Gbps. The USB 2.0 is black color coded and USB 3.0 is blue.

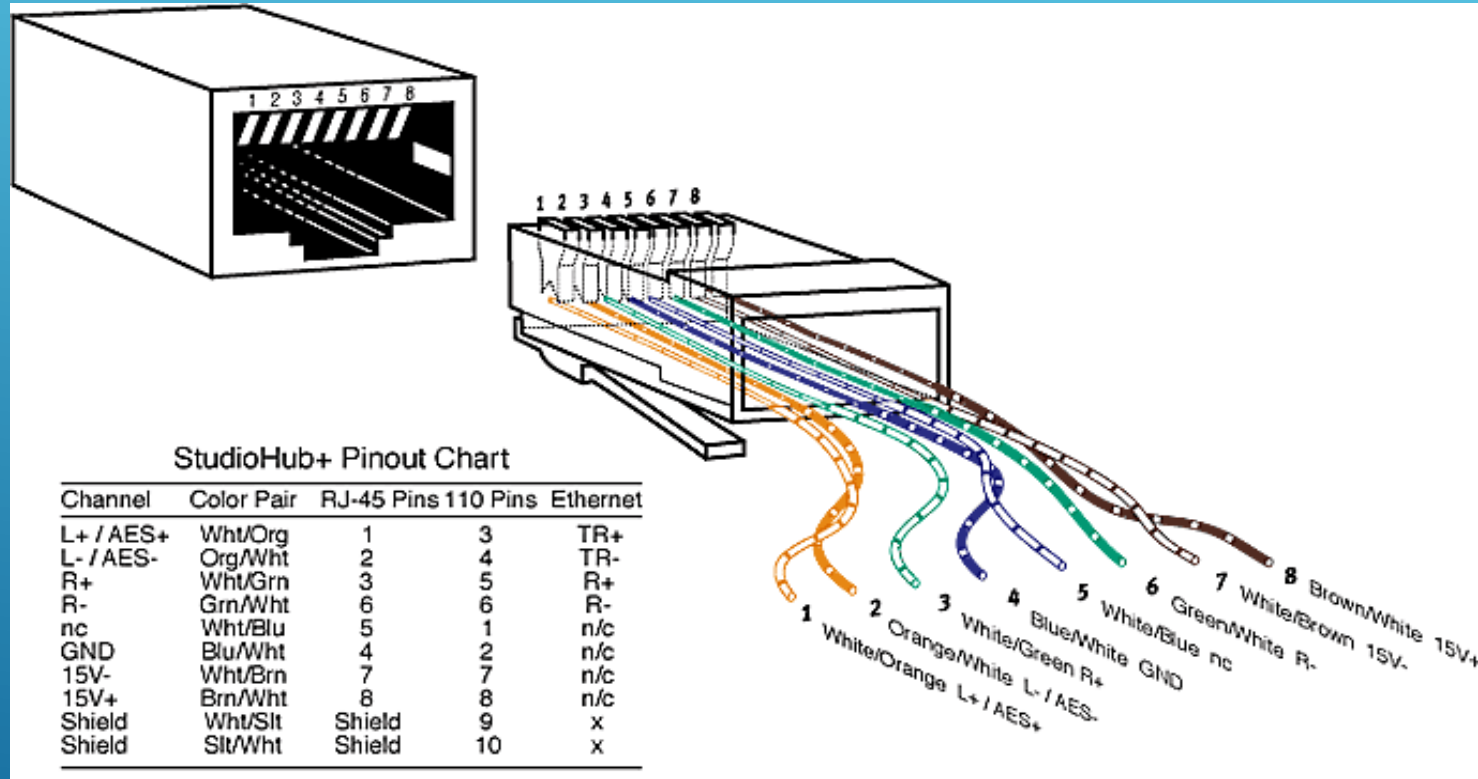
RJ-45



Ethernet is a networking technology that is used to connect your computer to internet and communicate with other computer or networking devices. The interface that is known as Registered Jack (RJ) and RJ – 45 port in particular is used for Ethernet over cable. RJ-45 connector is an 8 pin 0 8 contact (8P - 8C) type modular connector.

The un-keyed 8p – 8C modular connector is generally referred to the Ethernet RJ-45. Often, RJ-45 ports are equipped with two LEDs for indication transmission and packet detection.

Rj-45



RJ-11



Rj-11 is another type of Registered Jack that is used as an interface for telephone, modem or ADSL connections. RJ-45 and RJ11 ports look alike but RJ-11 is a smaller port and uses a 6 point – 4 contact (6P – 4C) connector even though a 6point – 2 contact (6P – 2C) sufficient.



E-SATA

e-SATA is an external ports capable of supporting both e-SATA and USB. Neither the SATA organization nor the USB organization has officially approved the e-SATA and must be used at user's risk.

POWER SUPPLY

A power supply is a device that supplies electric power to the components of the computer unit. Power Supply converts Alternating-Current(AC) Power into Direct-Current(DC).

AC- alternating current comes from the outlet

DC- direct current goes to the PC

Electrons – A flow of negatively charged particles
Called electrons

ELECTRICITY

Volts- 110-115v in USA other countries use 220-230v(pressure)

Amperes- The number of electron per second flowing between two
Points the amount of water per sec.

Watts- Amount of power (Typical PC today use about 500watts)

Ohms- Resistance Impedance

